4. Determine the number of permutations of the letters of the word SIMPLE if all are taken at a time?
5. A test consists of 10 multiple choice questions. In how many ways can the test be answered if
   (i) Each question has four choices?
   (ii) The first four questions have three choices and the remaining have five choices?
   (iii) Question number n has $n+1$ choices?
6. A student appears in an objective test which contain 5 multiple choice questions. Each question has four choices out of which one correct answer.
   (i) What is the maximum number of different answers can the students give?
   (ii) How will the answer change if each question may have more than one correct answers?
7. How many strings can be formed from the letters of the word ARTICLE, so that vowels occupy the even places?
8. 8 women and 6 men are standing in a line.
   (i) How many arrangements are possible if any individual can stand in any position?
   (ii) In how many arrangements will all 6 men be standing next to one another?
   (iii) In how many arrangements will no two men be standing next to one another?
9. Find the distinct permutations of the letters of the word MISSISSIPPI?
10. How many ways can the product $a^2b^3c^4$ be expressed without exponents?
11. In how many ways 4 mathematics books, 3 physics books, 2 chemistry books and 1 biology book can be arranged on a shelf so that all books of the same subjects are together.
12. In how many ways can the letters of the word SUCCESS be arranged so that all Ss are together?
13. A coin is tossed 8 times,
   (i) How many different sequences of heads and tails are possible?
   (ii) How many different sequences containing six heads and two tails are possible?
14. How many strings are there using the letters of the word INTERMEDIATE, if
   (i) The vowels and consonants are alternative (ii) All the vowels are together
   (iii) Vowels are never together (iv) No two vowels are together.
15. Each of the digits 1, 1, 2, 3, 3 and 4 is written on a separate card. The six cards are then laid out in a row to form a 6-digit number.
   (i) How many distinct 6-digit numbers are there?
   (ii) How many of these 6-digit numbers are even?
   (iii) How many of these 6-digit numbers are divisible by 4?
16. If the letters of the word GARDEN are permuted in all possible ways and the strings thus formed are arranged in the dictionary order, then find the ranks of the words (i) GARDEN (ii) DANGER.
17. Find the number of strings that can be made using all letters of the word THING. If these words are written as in a dictionary, what will be the 85th string?
18. If the letters of the word FUNNY are permuted in all possible ways and the strings thus formed are arranged in the dictionary order, find the rank of the word FUNNY.
19. Find the sum of all 4-digit numbers that can be formed using digits 1, 2, 3, 4, and 5 repetitions not allowed?
20. Find the sum of all 4-digit numbers that can be formed using digits 0, 2, 5, 7, 8 without repetition?

4.5 Combinations

Let us suppose there are four persons A, B, C and D (actual names may be used here) and we have to select three of them to be a part of a committee. In how many ways can we make this selection? For example, A, B, C is one possible choice. Here the order of selection is immaterial. Thus A, B, C is the same as B, A, C or C, A, B as long as the same three persons are selected. Thus the possible